



What We Have Built

Product vision, features & architecture

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01

Product Vision

The problem we are solving, who we are solving it for, and the value we deliver to users and institutions.

The Problem

Latin American individuals with mid-to-high incomes increasingly manage finances across multiple bank accounts, fintech platforms, cards, shared household expenses, subscriptions, and currencies. No single tool gives them a unified, automated, real-time picture of their financial life. Managing personal finances today means juggling bank apps, spreadsheets, Splitwise for shared expenses, and separate budgeting tools — none of which talk to each other.

On the institutional side, Latin American fintechs and financial institutions have limited access to structured, high-quality individual financial behavior data. This constrains their ability to assess creditworthiness and design personalized financial products for a market that is largely underserved by traditional credit infrastructure.

Who We Are Building For

TARGET USER

Latin American individuals & couples

TARGET CUSTOMER (B2B)

LATAM fintechs & financial institutions

The target user is a Latin American banked or semi-banked individual with any level of financial knowledge who wants to track, understand, and optimize their personal finances. The product must be simple enough for a first-time budgeting app user, yet powerful enough for someone managing a household with a partner, multiple accounts, and foreign currency income.

The B2B customer is any Latin American fintech or financial institution seeking behavioral and creditworthiness signals on individuals — and over

time, consumer goods and leisure companies that benefit from understanding individual financial patterns.

Value We Deliver

To the User

A one-stop automated platform to track, understand, analyze, and act on all personal finances. Every income and expense transaction is AI-classified and delivered as a real-time alert via WhatsApp the moment it happens. Rather than replacing apps like Splitwise, Rocket Money, or bank apps, Luka connects and displays everything in a single place. Predictive budgeting and personalized best practices help users achieve specific financial goals based on their own behavior.

To the Customer

Structured and detailed individual financial behavior data — aggregated or individual — to make creditworthiness predictions and offer personalized financial products to current or new customers. A centralized channel to access a pool of creditworthy individuals who previously had no relationship with their institution.

Business Model

Luka operates a two-sided revenue model, similar to YouTube's approach to advertising. Users pay a small subscription that keeps their individual data private. B2B customers pay for access to aggregated, anonymized behavioral data. The user always comes first: a large, satisfied user base generating behavioral data is the foundation that makes the institutional product valuable.

Geographic Scope

Luka is LATAM-first. The platform currently supports Chile, Colombia, Mexico, Peru, Brazil, and the United States, with a three-layer email parsing engine covering 101 banks across the six countries. All transaction capture, parsing, classification, and display logic supports multiple currencies — nothing is hardcoded to a single country or currency.

02

The Application

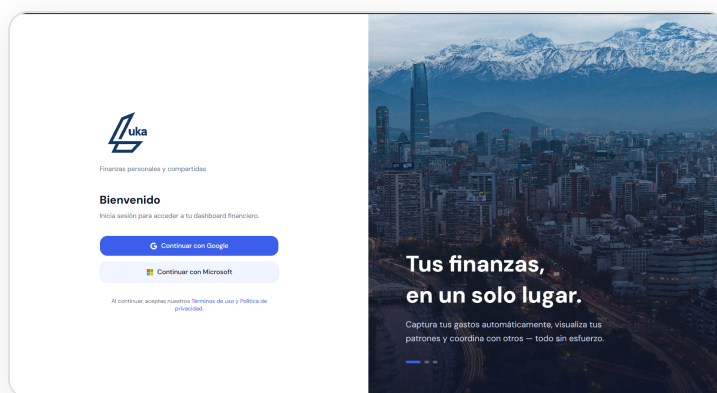
A tour of every screen in Luka — what users see, what they can do, and how it fits into their daily financial life.

Authentication & Onboarding

LOGIN

Sign In

Users sign in with Google for Gmail-connected accounts or Microsoft for Outlook-connected accounts. Supabase handles the full OAuth flow. First-time users are automatically provisioned and guided through a three-step onboarding sequence: household setup (individual or couple), bank connection, and WhatsApp phone number verification. The login screen is bilingual and fully responsive on mobile and desktop.



Luka — Login screen (luka-lovat.vercel.app)

Dashboard

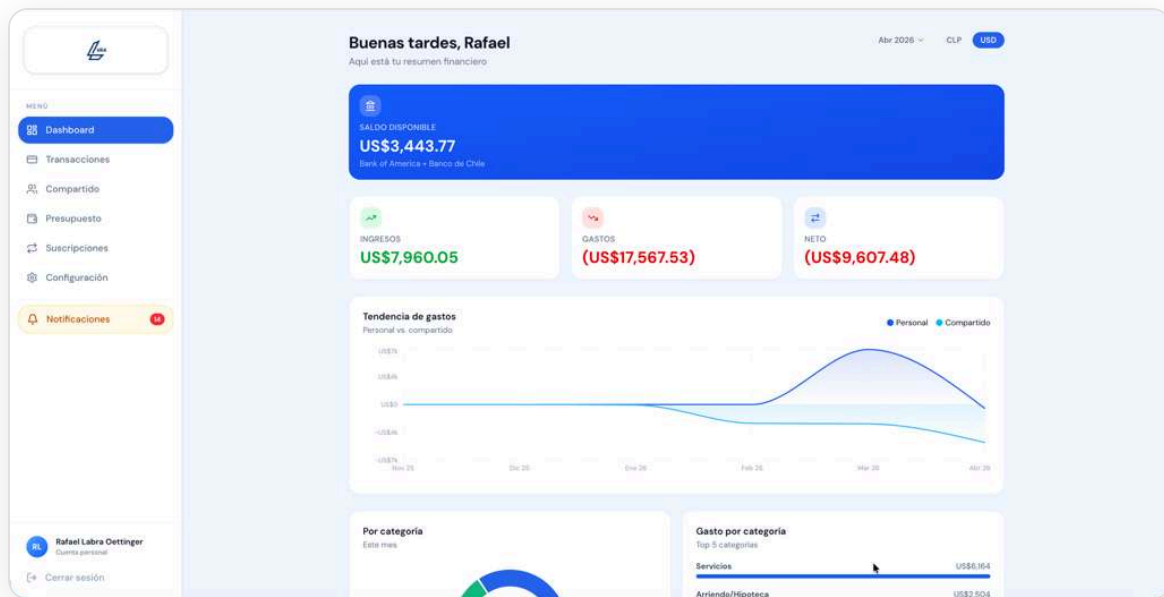
HOME

Financial Summary

The home screen is the daily command center. A time-sensitive greeting at the top is paired with a month selector and a currency toggle — users with multiple currencies switch the entire view instantly. A balance card shows live checking account balances across all connected accounts. Three cash flow summary

cards display total income, total expenses, and net position for the selected month and currency.

Below the summary row, a six-month spending trend chart plots personal and shared expenses as separate area series. A category donut chart shows the top five spending categories for the month, and a budget bar chart shows progress against any per-category caps. The page closes with the five most recent transactions and a link to the full history. All widgets update in real time as new transactions arrive.



Home — balance card, cash flow summary, spending trend chart, category donut

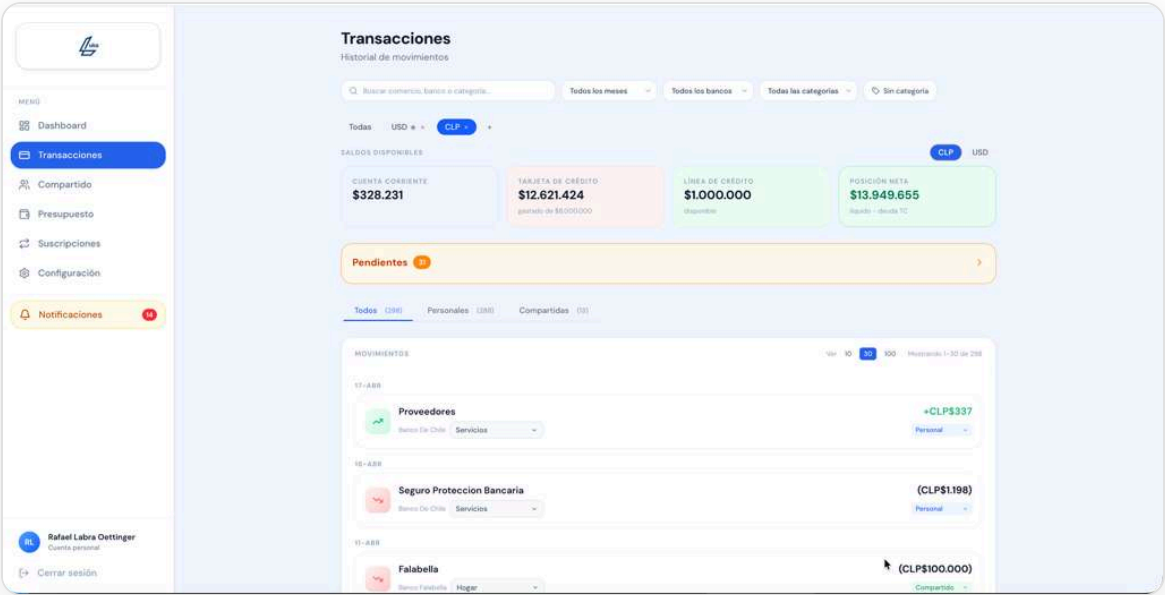
Transactions

TRANSACTIONS

Full Transaction History

The transactions page gives users complete control over their history. At the top, account balance summary cards show checking balance, credit card usage versus limit, line of credit balance, and net position — filtered by the selected currency. A currency pill bar lets users switch between CLP, USD, and any other currency they have configured, with an inline option to add new ones.

A pending block surfaces transactions awaiting reconciliation or classification. Below it, three tabs separate all, personal, and shared household transactions. Each tab shows a paginated table — 10, 30, or 100 rows — with filters for month, bank, category, free-text search, and an uncategorized-only toggle. Every row is editable inline: category and split type can be changed directly from the list.



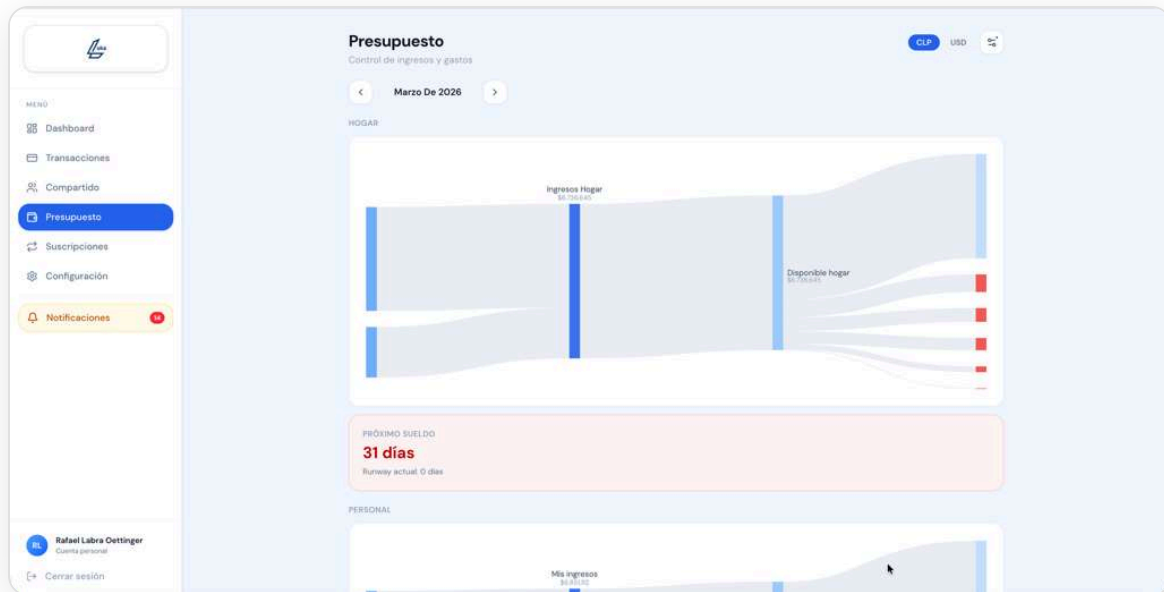
Transactions — balance summary, currency pills, pending block, paginated transaction table

Budgets

BUDGETS Sankey Flow & Budget Configuration

The budgets page visualizes how household income flows through allocations using a multi-level Sankey diagram. Income sources appear on the left, a central household income hub aggregates them, and allocation nodes on the right break out fixed bills, installment payments (cuotas), savings target, personal allocation, and discretionary spending — each fanning out into individual spending categories. Users with a partner see their own income broken out while the partner appears as a single aggregated node, preserving financial privacy by design.

A gear icon opens the budget configuration modal, consolidating all budget settings: savings target, personal spending allocation, payday day, household contribution mode, and per-category spending caps. A runway card estimates how many days the current balance would last at the current burn rate.

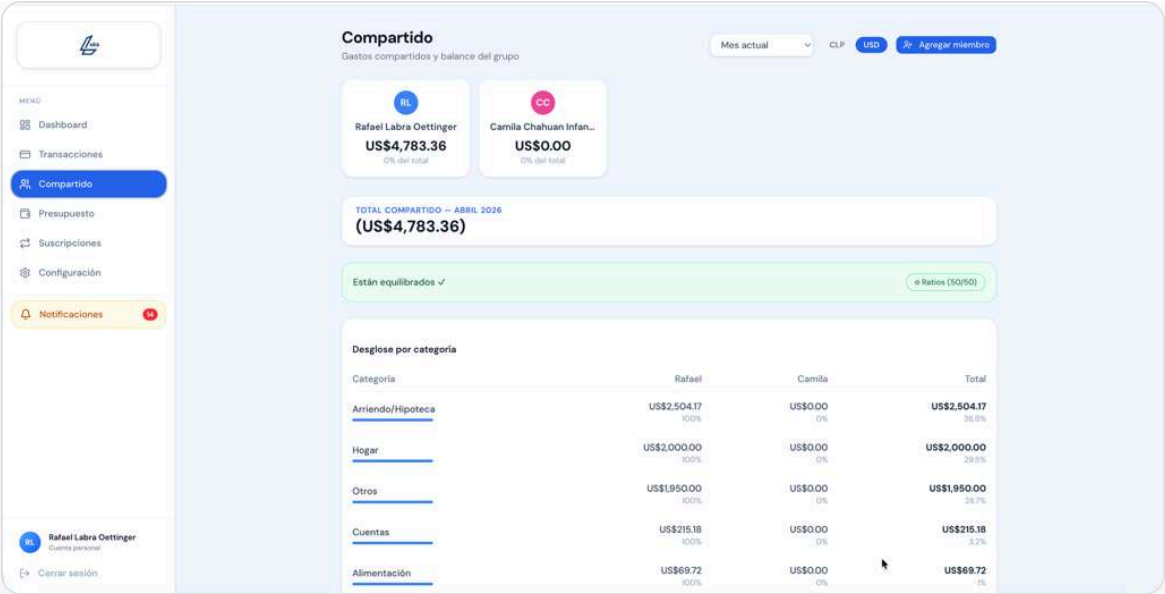


Budgets — Sankey income-to-spending flow, hogar and personal views, runway card

Household

HOUSEHOLD Shared Finance Management

The household page is built for couples and groups. Member cards show each person's total income, total expenses, and net contribution for the selected month. A category breakdown table shows spending by category across the entire household with each member's share. A settlement card calculates who owes whom at month-end based on the agreed contribution ratio. The split ratio is adjustable from a settings modal, and a new partner can be invited via a shareable link generated directly from this page.



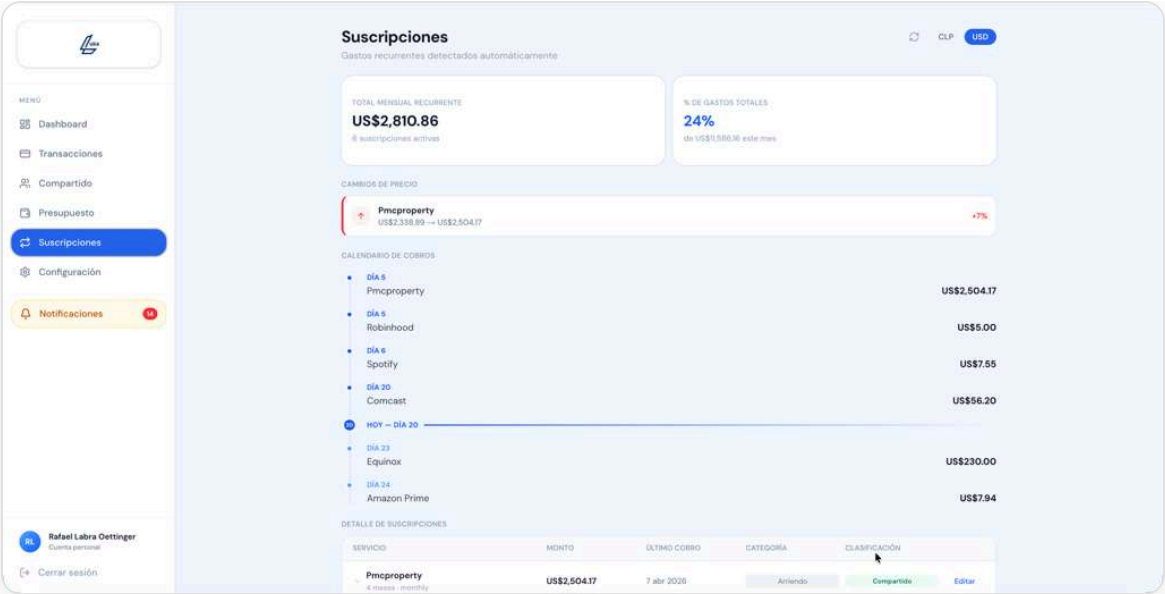
Household — member contribution cards, category breakdown by member, settlement status

Subscriptions

SUBSCRIPTIONS

Recurring Expense Detection

Luka automatically detects recurring charges by analyzing transaction patterns across two or more consecutive months with a plus-or-minus 20% amount tolerance. The subscriptions page lists every detected subscription with its merchant, amount, last charge date, estimated next charge date, and current status. A billing calendar shows upcoming charges by day of month. Each subscription has a Personal or Shared toggle — flipping it reclassifies the last three months of underlying transaction splits automatically so the household budget reflects the change immediately.



Subscriptions — total recurring spend, billing calendar, subscription detail table with personal/shared toggle

Notifications

NOTIFICATIONS

In-App Alert Center

The notifications page shows a chronological list of all in-app alerts: new transaction detected, reconciliation completed, budget cap reached, and system messages. Each notification can be marked as read or deleted individually. An unread badge on the navigation icon updates every 30 seconds. WhatsApp transaction alert preferences are controlled from the notification preferences toggle in settings.

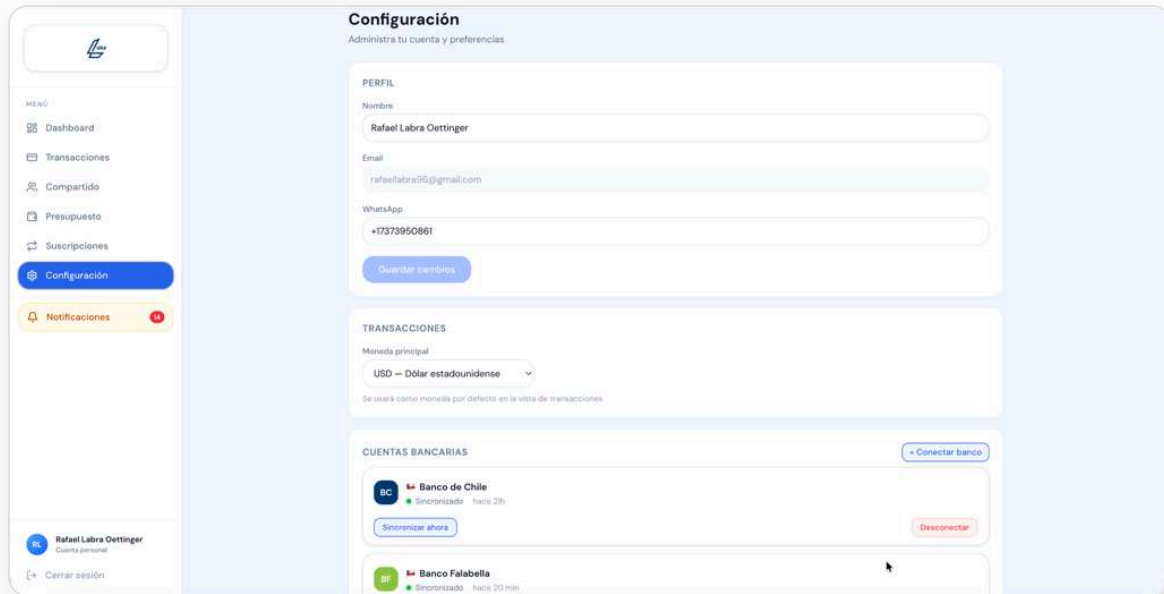
Settings

SETTINGS

Account & Preferences

Settings is organized into collapsible sections. Profile covers display name, email (from OAuth, read-only), and WhatsApp number with a re-verification flow. Transactions config sets the primary currency. Bank accounts lists all

connected accounts — from Luka Connect scraping, Plaid, or manually added — showing sync status, with options to sync or disconnect. Household controls contribution mode and split ratio. Categories lets users reorder spending categories with drag-and-drop, hide unused ones, and add custom categories. A privacy section links to the data deletion flow.



Settings — profile, primary currency, connected bank accounts (Banco de Chile, Banco Falabella)

WhatsApp Integration

WhatsApp is the primary notification and action channel, bringing Luka to the messaging app users already have open all day. When a new transaction is detected, Luka sends an interactive message with the merchant name, amount, bank, and timestamp. For expenses and income, the message includes action buttons — Personal and Shared — followed by an inline category picker. The user taps twice to classify and confirm without ever opening the web app.

Transfers between own accounts — credit card bill payments, checking-to-savings moves — receive an informational-only message with no action required. Users can also type a free-text message to Luka's WhatsApp number

to log a manual expense in natural language, which is parsed by AI and saved as a transaction.

03

Architecture

How the system is built — the infrastructure, data pipeline, and technical decisions behind the product.

System Overview

Luka is a monorepo with three deployed backend services — a FastAPI API server, a fast ARQ worker, and a slow ARQ worker — all sharing a Supabase PostgreSQL database and a Redis instance. The frontend is a Next.js application deployed on Vercel.



Backend

The API is built with Python 3.12 and FastAPI, exposing 16 routers covering all product domains. JWT tokens are validated locally using PyJWT and JWKS — no round-trip to Supabase on every request. User profiles are cached in Redis for five minutes. SQLAlchemy 2.0 async handles all database access through asyncpg.

Email Parsing Pipeline

Transaction capture from bank email notifications uses a three-layer parser. The first layer executes declarative JSON templates at zero LLM cost — generated and maintained automatically by a nightly Template Agent. The second layer runs a Gemini LLM waterfall through four models escalating by confidence threshold, with a circuit breaker that cuts off LLM calls if the error rate exceeds 50% in a 20-request window. The third layer is a regex fallback for known legacy bank formats. Every parsed email is logged for audit and template training.

The Template Agent runs nightly at 2 AM. It re-parses a 25% sample of recent emails to validate active templates — any amount mismatch triggers automatic template retirement. It also discovers banks with enough data to generate a new template, validates it against LLM ground truth, and promotes it to production automatically, continuously shifting cost from LLM to templates as data accumulates.

Worker Architecture

Two separate ARQ worker processes prevent slow jobs from blocking real-time email processing. The fast worker handles email processing and all cron jobs, supporting up to 20 concurrent jobs with a 60-second timeout. The slow worker handles bank syncs, LLM merchant review, reconciliation, and the Template Agent, supporting up to 5 concurrent jobs with a 10-minute timeout.

Frontend

Next.js 16 with App Router, React 19, Tailwind CSS 4, and shadcn/ui. Zustand 5 manages client state persisted to localStorage; TanStack Query 5 manages all

server state with a 30-second stale time and eight queries prefetched in parallel on app load. The Recharts charting library is deferred via dynamic imports to avoid blocking initial paint. The app is a PWA with a 30-minute inactivity timeout.

Bank Connections & Reconciliation

LATAM banks connect via Luka Connect, a standalone scraper covering nine Chilean banks with AES-256-GCM encrypted credential storage and six-hour sync cadence. US banks connect via Plaid with cursor-paginated transaction sync and smart mapping logic distinguishing person-to-person payments from own-account transfers. A three-tier reconciliation engine — exact, fuzzy, and sum-match — runs both inline during sync and as a daily cron job, enriching bank-sourced transactions with user-edited categories from email-sourced transactions and eliminating duplicates.

Database

Supabase PostgreSQL 15 with PgBouncer connection pooling, managed with Alembic across 37 migration versions. Row-level security is enabled on sensitive tables. A SECURITY DEFINER aggregate RPC is used for partner contribution data, ensuring no raw partner transaction rows are ever exposed across household members — privacy is enforced structurally, not just by application logic.